

HOMEWORK 2

1. Let G be a group and $a, b \in G$.

(a) Prove that $a^n \cdot a^m = a^{n+m}$ and $(a^n)^m = a^{nm}$ for all $n, m \in \mathbb{Z}$.

(b) Prove that $\text{ord}(a^n) = \frac{\text{ord}(a)}{\gcd\{n, \text{ord}(a)\}}$ if $\text{ord}(a) < \infty$.

(c) Prove that $\text{ord}(ab) = \text{ord}(a) \cdot \text{ord}(b)$ if a and b commute and $\gcd\{\text{ord}(a), \text{ord}(b)\} = 1$.

2. Let $H \subset G$ be a subgroup. Show that the correspondence $Ha \mapsto (Ha)^{-1} = a^{-1}H$ is a bijection between the sets of right and left cosets of H in G .

3. Let $H \subset G$ be a subgroup. Suppose that for any $a \in G$ there exists $b \in G$ such that $aH = Hb$. Show that H is normal in G .

4. Let G be the smallest subgroup of $GL_2(\mathbb{C})$ containing $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

(a) Prove that G is a non-abelian group of order 8.

(b) Determine all subgroups of G and prove that they are all normal in G .

5. Let N be a subgroup in the center $Z(G)$ of G . Show that N is normal in G . Prove that if the factor group G/N is cyclic, then G is abelian.

6. Prove that if a group G contains a subgroup H of finite index n , then G contains a normal subgroup N of finite index dividing $n!$ such that $N \subset H$. (Hint: Consider the homomorphism of G to the symmetric group of all left cosets of H in G taking any $x \in G$ to f_x defined by $f_x(aH) = xaH$.)

7. Let N be a normal subgroup of a group G . Prove that the assignment $H \mapsto H/N$ establishes

(i) A bijection between the set of all subgroup of G containing N and all subgroups of G/N ;

(ii) A bijection between the set of all normal subgroup of G containing N and all normal subgroups of G/N .

Construct the inverse bijections using the canonical homomorphism $\pi : G \rightarrow G/N$.

8. (a) Show that the group $\text{Inn}(G)$ of all inner automorphisms of a group G (given by $a \mapsto gag^{-1}$ for some $g \in G$) is a normal subgroup in the group $\text{Aut}(G)$ of all automorphisms of G .

(b) Find all automorphisms of all (finite and infinite) cyclic groups.

9. Let x and x' be two elements in the same orbit under some action of a group G on a set. Show that the stabilizers $G(x)$ and $G(x')$ are conjugate in G .

10. Let a group G act on two sets X and Y . We say that X and Y are *G -isomorphic* if there is a bijection $f : X \rightarrow Y$ such that $f(gx) = g(f(x))$ for every $x \in X$ and $g \in G$. Prove that if G acts on X transitively, then X is G -isomorphic to the set of left cosets G/H for some subgroup $H \subset G$ (with the action of G on G/H by left translations).